Future-Event Minimal Quotients for a Circular Registered Absorption System

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Abstract

We give an exact, observation-contract-dependent state minimization with explicit distinguishing certificates for a circular packet system generated by rotation and one adjacent collapse. Starting from identity-placed singletons, we register the fusion that creates the source pair $F = \{0, 1\}$. The full-summary observer reads the current placement and the registered absorption history; the future-event observer reads only the complete selected events emitted by subsequent letters. These contracts induce different right congruences on the same reachable state space.

For every $n \geq 3$, we classify the future-event congruence: all states from which no selected event remains possible form one class; each registerable pre-registration placement is a class; and registered nonterminal states are equivalent exactly when their current placed partitions agree. Explicit raw words distinguish every inequivalent pair within $6n - 10$ letters. The proof combines an outside-anchored sweep for registered placements with an atomwise gap probe whose entire coalescence fiber is characterized.

Writing R_n for registerable pre-registration placements, A_n for registered nonterminal placements, and s_n for complete registered summaries, the two minimal state counts are

$$Q_n^{\text{future}} = 1 + R_n + A_n, \qquad Q_n^{\text{full}} = 1 + R_n + s_n.$$

Their exponential growth bases are 4 and 9/2, respectively. These are model- and observer-relative state counts, not reset thresholds or generic memory lower bounds.

Keywords: Mealy right congruence, circular packet dynamics, event registration, state minimization, constructive distinguishing word

1 Introduction

A state sufficient to continue an execution need not retain every event that led to it. Conversely, whether an old event can be forgotten depends on the observer. The event-anchored model of (Chen, 2026b) follows one selected fusion and its subsequent carrier. A separate counting result (Chen, 2026a) enumerates its complete registered summaries for the circular adjacent-collapse action. Neither count is automatically the number of states needed to predict only *future* selected events.

Here we fix the same raw action and registered pair, but compare two precise observation contracts. A Moore observer displays the complete current registered summary. A Mealy observer emits, letter by letter, only newly occurring selected registration or absorption records. Right indistinguishability for these contracts follows the classical sequential machine viewpoint (Mealy, 1955; Moore, 1956; Nerode, 1958). Our contribution is an explicit classification and count of the classes of these two minimal right congruences, together with constructive distinguishing words that establish minimality independently of finite enumeration.

The key asymmetry is that a registered selected chain is never emitted again. Once registered, its current placement determines all future selected events. Already for $n = 5$, the two actual words

$$w_1 = d\,p^3 d\,(pdp^4)p^2 d,$$
$$w_2 = d\,(p^4 dp)d\,(p^3 dp^2)(p^4 dp)d$$

both reach $P(0) = \{0, 1, 2, 3\}$ and $P(4) = \{4\}$. Their registration origins agree, but w_1 first assembles $\{2, 3\}$ away from the carrier and then absorbs it in one selected event, whereas w_2 absorbs $\{2\}$ and $\{3\}$ in two selected events. The full-summary observer separates these old chains immediately; every future selected-event trace agrees. Before registration, however, distinct placements remain distinguishable. A gap probe can coalesce with a different starting placement, so its target must be chosen carefully. An exact description of that probe's coalescence fiber yields a strictly decreasing gap and resolves the pairwise separation problem.

The mathematical scope is fixed throughout: $n \geq 3$, the identity singleton seed on $Q = \mathbb{Z}/n\mathbb{Z}$, raw letters p and d, the ordered collapse ports $(0, 1)$, selected source pair $F = \{0, 1\}$, complete event records, all reachable histories, and no word-length budget. Section 2 fixes the two contracts. Sections 3–5 prove placement control and the two probe lemmas. Section 6 proves the quotient and depth theorems. Section 7 counts the classes and the explicit probe-coalescence graph.

2 Model and observation contracts

Let Ω be a set of n source atoms and $Q = \mathbb{Z}/n\mathbb{Z}$ a set of physical sites. We identify their labels only at the singleton seed $P_0(q) = \{q\}$. A **placed partition** P assigns to each occupied site $q \in Q$ a nonempty packet $P(q) \subseteq \Omega$. Its packets are disjoint and cover Ω; unoccupied sites are holes.

The raw letter p acts by $p(q) = q + 1$. The letter d acts by

$$d(1) = 0, \qquad d(q) = q \quad (q \neq 1).$$

Packets follow the site maps and packets with the same image merge by union. Each letter therefore defines a total deterministic update of placed partitions. A word is executed chronologically, from left to right.

We select $F = \{0, 1\} \subseteq \Omega$. A **registration** occurs when an actual d-fusion first creates a packet whose source set is exactly F. Its record contains the letter, both ordered kernel ports, the two actual parent packets, and the collision site. Since packets never split, a registration can occur at most once. After registration, a **selected absorption** is an actual fusion involving the packet containing F; its record again retains the ordered parent packets and ports. Unrelated fusions may occur, but emit no selected event.

The retained execution state is $x = (P, o, A)$: current placed partition, registration origin o (or $\perp$ before registration), and the ordered selected absorption chain A. Let X_n be the set of such states reachable from $(P_0, \perp, ())$ by actual p, d words. The selected event output $\epsilon(x, a)$ is null unless letter a creates the registration or a selected absorption, when it is the complete new record. For $w = a_1 \cdots a_m$, let $\mathrm{Trace}(x, w)$ be the length-m sequence of these outputs. Define the future-event right congruence

$$x \equiv_{\mathrm{future}} y \quad \Longleftrightarrow \quad \mathrm{Trace}(x, w) = \mathrm{Trace}(y, w) \text{ for every } w \in \{p, d\}^*. \tag{2.1}$$

The **full-summary** observer instead displays $\perp$ before registration and the entire current (P, F, o, A) afterwards. Its Moore right congruence $\equiv_{\mathrm{full}}$ compares the displayed output after

every common word, including the empty word. Write Q_n^{future} and Q_n^{full} for the numbers of equivalence classes on X_n.

There are three useful semantic regions. Let $\mathcal{R}_n$ consist of unregistered states from which registration of F remains possible. Let $\mathcal{Z}_n$ consist of the other unregistered states together with registered rank-one states. Finally, registered states with at least one packet outside the F-carrier are called **registered nonterminal**. Let $R_n = |\mathcal{R}_n|$, let A_n count their distinct current placed partitions, and let s_n count *all* registered complete summaries, including rank one.

3 Source order, placement control, and counts

Lemma 3.1 (source order and nonfusion control). Every reachable packet is a consecutive source arc, and the occupied packets occur in the positive cyclic source order. If two placed partitions have the same packet blocks in that order, one is reachable from the other by an actual word with no fusion.

Proof. Both statements hold at the seed. Rotation preserves packet order. A d-fusion joins packets at physical sites $0, 1$, adjacent in the occupied cyclic order, so it joins consecutive source arcs without changing the order of the remaining packets. If site 0 is empty and site 1 occupied, d instead moves that packet into the hole at 0 without fusion or a change of cyclic packet order; the other nonfusion cases leave the order unchanged. For placement control, use the fusion-free placement-control lemma of (Chen, 2026a, Lemma 2.2). Its elementary guarded move is

$$E_j = p^{[-j]_n} d p^{[j]_n}.$$

When site j is a hole and $j+1$ is occupied, this actual word moves only that packet from $j+1$ to j at its end; its middle d does not fuse. Let r be the number of blocks. If $r = 1$, a rotation aligns the sole block even when holes remain. Assume $r \geq 2$ for the gap transfers. Name the blocks in their common cyclic order $B_0, \ldots, B_{r-1}$ and let h_i count the holes after B_i and before B_{i+1}, cyclically. The total $\sum_i h_i = n - r$ is fixed. Applying E_j at the hole immediately before B_{i+1}, when $h_i > 0$, changes exactly

$$(h_i, h_{i+1}) \longmapsto (h_i - 1, h_{i+1} + 1) \qquad (r \geq 2,\ h_i > 0).$$

The packet crosses that hole but no other packet, so the block order and every packet's source set are unchanged. If g is the target gap vector, choose e with $h_e > g_e$ and f with $h_f < g_f$. Route one hole from gap e to gap f through consecutive gap indices. The first transfer is legal because $h_e > 0$; each subsequent donor has just received that hole. The intermediate gaps have zero net change, so this route replaces h by $h - e_e + e_f$ and decreases $\sum_i (h_i - g_i)_+$ by one. Finitely many actual routes reach g. A final power of p aligns B_0 and therefore all blocks. If there are no holes, rotation alone suffices. At each crossing the guard is checked on the concrete state produced by the preceding word. □

Lemma 3.2 (registration criterion). An unregistered reachable state is in $\mathcal{R}_n$ if and only if source atoms 0 and 1 remain separate singleton packets. In such a state, if their sites are a, b, then the positive arc from a to b contains only holes between them and $1 \leq [b - a]_n \leq n - 2$.

Proof. If either atom has already fused with an outside atom, no future fusion can create exactly F; if they had fused with each other, that fusion would already have registered F. Conversely, when both are singleton packets, they are consecutive in source order by Lemma 3.1. Nonfusion placement control puts them at ports $0, 1$ in that order, after which one d

registers F. An outside packet remains because $n \geq 3$, giving a nonempty complementary arc and the stated gap range. □

Proposition 3.3 (placement counts). For $n \geq 3$,

$$\begin{aligned} R_n &= n\binom{2n-4}{n-3}, \\ A_n &= n\sum_{k=1}^{n-2}(n-1-k)\binom{n+k-2}{k} \\ &= 2(2n-3)\binom{2n-4}{n-2} - n(n-1). \end{aligned} \tag{3.1}$$

Proof. Partition the $n-2$ outside atoms into $t \geq 1$ nonempty consecutive packets. There are $\binom{n-3}{t-1}$ cuts. The $t+2$ named packets have $(t+2)\binom{n}{t+2} = n\binom{n-1}{t+1}$ source-ordered physical placements. These are actually reachable without registering F: assemble each outside interval from singleton packets by adjacent fusions, keeping sources $0, 1$ separate, and use Lemma 3.1 to place the result. Thus

$$R_n = n\sum_{t=1}^{n-2}\binom{n-3}{t-1}\binom{n-1}{t+1} = n\binom{2n-4}{n-3}.$$

For the second count, fix a proper source-arc carrier containing F and leave $k \geq 1$ outside atoms. Its left and right extensions from F have sum $n-2-k$, so there are $n-1-k$ such carriers. For each, partition the k outside atoms into t nonempty consecutive packets and place those t packets and the carrier in cyclic order. Summing over t gives $n\binom{n+k-2}{k}$ placements. They are actual registered placements: register F, extend its carrier by adjacent singleton absorptions to the fixed source arc, assemble outside packets by fusions not involving the carrier, then use Lemma 3.1 to reach any target placement. This is also a direct consequence of the constructive fixed-carrier fiber theorem in (Chen, 2026a); only placement existence, not its history count, is used here. The carrier is read from the packet containing F, so different carriers do not overlap in the count.

For the closed form, put $m = n-2$ and use $\sum_{k=0}^{m}\binom{m+k}{k} = \binom{2m+1}{m}$ and $\sum_{k=0}^{m} k\binom{m+k}{k} = (m+1)\binom{2m+1}{m-1}$. Substitution and elementary binomial ratios give (3.1). □

4 Registered-placement probes

For a site q, define the raw word

$$W_q = p^{[-q]_n} d(pd)^{n-2}, \qquad |W_q| \leq 3n-4. \tag{4.1}$$

The rotation in this word is an actual nonnegative power of p.

Lemma 4.1 (aligned sweep). If two registered nonterminal states have their F-carriers at the same site q but different placed partitions, W_q gives different future-event traces.

Proof. Rotate the common carrier to site 0. Name the packet initially at site j by B_j, using an empty symbol at holes. The first d records B_1 if present and leaves the carrier at 0. In the ith following pd pair, the p moves the carrier to 1 and brings the packet initially at site $n-i$ to 0; the d records that packet if present and returns the carrier to 0. The output positions therefore read

$$B_1, B_{n-1}, B_{n-2}, \ldots, B_2$$

with null exactly at holes. The first nonnull record also gives the original carrier packet. Such a record exists because the state is nonterminal. Hence the complete trace recovers the normalized placed partition and separates different placements. □

Lemma 4.2 (first-event orientation decoder). Fix a site q and a registered nonterminal state z. After the initial rotation $p^{[-q]_n}$ in W_q, let r be the carrier site. If the first selected event in the remaining word $d(pd)^{n-2}$ occurs at its initial d with the carrier as right parent, then $r = 1$. If it occurs at the d of the kth pd pair with the carrier as left parent, then $r = n-k$. An event at the initial d with the carrier as left parent instead gives $r = 0$.

Proof. An event at the initial d with the carrier as right parent puts it at port 1; the left-parent case puts it at port 0. For a first event with the carrier as left parent in the kth pd pair, consider all earlier d letters. If one had left the carrier at site 0, the next p would put it at site 1. Each subsequent nonselected d would move it back to 0, while any selected d would use it as the right parent. The same applies if an earlier nonselected d moved the carrier from site 1 into an empty site 0. Neither history permits the stated first left-parent event. Thus the carrier has been fixed by each earlier d and moved only by the k rotations in the pd pairs. It reaches port 0 when $r + k = 0$ modulo n, giving $r = n - k$. These are implications for the specified first-event types; no assertion is made about traces with no selected event. □

Lemma 4.3 (outside-anchored phase). Let x be registered nonterminal, with carrier site c, and choose the site $q \neq c$ of any outside packet. For every registered nonterminal comparison state y,

$$\mathrm{Trace}(x, W_q) = \mathrm{Trace}(y, W_q) \quad \Longrightarrow \quad \mathrm{carrierSite}(y) = c. \tag{4.2}$$

Proof. After the initial rotation, the outside anchor of x is at 0. Put $j = [q - c]_n$, so the carrier begins at $r = [c - q]_n = n - j$. If $j = n - 1$, the first d has the carrier at port 1 and the anchor at port 0, producing an immediate selected event with the carrier as right parent. If $1 \leq j \leq n - 2$, the first d leaves the anchor packet at site 0, possibly after an unrelated fusion. Until it meets the carrier, each following pd pair moves that packet from 0 to 1 and back to 0; it may absorb further outside packets but cannot lose its anchored source atoms. The first selected event is the d in the jth pd pair: the carrier reaches port 0 and the anchor is at port 1, so the carrier is left parent.

Offset $j = [q - c]_n$	First selected event	Carrier parent	Initial normalized site r
$n - 1$	Initial d	Right, at port 1	1
$1 \leq j \leq n - 2$	d in the jth pd pair	Left, at port 0	$n - j$

The same first-event position and parent role occur in y when the complete traces agree. Lemma 4.2 decodes its initial normalized carrier site without assuming that its packet at site 0 matches the anchor packet of x. Both normalized carrier sites are therefore r; undoing the common rotation gives the claim. □

Corollary 4.4 (registered separation). Every two registered nonterminal states with different current placements are separated by one W_q. If the carrier sites differ, choose an outside site q of the first state and apply Lemma 4.3. If they agree, choose q to be the common carrier site and apply Lemma 4.1. In either case the same raw word is run from both states.

5 Pre-registration gap probes

For a placed partition P, let $\mathrm{site}_P(z)$ be the site of the packet containing source atom z. For a chronological word $w = a_1 \cdots a_m$ let $\phi_w = f_{a_m} \circ \cdots \circ f_{a_1}$, where $f_p = p$ and $f_d = d$.

Lemma 5.1 (atomwise execution). For any actual placed partition, word w, and source atom z,

$$\mathrm{site}_{\delta_w(P)}(z) = \phi_w(\mathrm{site}_P(z)). \tag{5.1}$$

Furthermore, the entire final placed partition is recovered from these sites: its packet at q is the nonempty fiber $\{z : \mathrm{site}_P(z) = q\}$.

Proof. Under one letter every atom in a packet moves to the image of its packet's site. When images coincide, their atom sets merge at that image. This proves (5.1) for a letter even through fusion. Induction on the word proves the formula, and the fiber identity recovers the partition. □

For $x \in \mathcal{R}_n$, let a, b be the sites of singleton sources $0, 1$ and set $g = [b - a]_n$. By Lemma 3.2, $1 \le g \le n - 2$ and sites $a + 1, \ldots, a + g - 1$ are holes. Define

$$V_x = p^{[1-b]_n}(dp)^{g-1}d, \qquad |V_x| \le 3n - 6. \tag{5.2}$$

Lemma 5.2 (site fibers and first registration). Write $V_x = v_x d$ and let $\Phi_x = \phi_{v_x}$. Then

$$\Phi_x^{-1}(0) = \{a\}, \qquad \Phi_x^{-1}(1) = \{a + 1, \ldots, a + g\}. \tag{5.3}$$

Φ_x is injective on the complement of $\{a, a + 1, \ldots, a + g\}$ and maps that complement away from ports $0, 1$. The word V_x first registers exactly F on its last letter.

Proof. The initial rotation sends a to $1 - g$ and b to 1. One chronological dp block has site map $T(0) = T(1) = 1$ and $T(j) = j + 1$ otherwise. Use integer representatives for the interval $\{1 - k, \ldots, 1\}$ and reduce their sites modulo n. For $0 \le k \le g - 1 \le n - 3$, the stronger induction statement is

$$T^k(j) = \begin{cases} 0, & j = -k, \\ 1, & j \in \{1 - k, \ldots, 1\}, \\ j + k, & j \notin \{-k, 1 - k, \ldots, 1\}. \end{cases} \tag{5.3a}$$

At $k = 0$ this is the identity map. Suppose it holds at k. The old fiber over 1 stays there, while its neighboring site $-k$, whose image was 0, also moves to 1 under the next T. The site $-(k + 1)$ previously mapped to -1, and now maps to 0. Every other site is still translated by one. The displayed cases are disjoint in the stated range, so this proves the induction and, in particular,

$$(T^k)^{-1}(0) = \{-k\}, \qquad (T^k)^{-1}(1) = \{1 - k, \ldots, 1\};$$

the remaining nonempty fibers are singletons. With $k = g - 1$, the initial rotation takes a to $-k$ and takes the sites $a + 1, \ldots, a + g$ to $1 - k, \ldots, 1$, proving (5.3). The source-0 atom thus reaches port 0, and source 1 reaches port 1. The open arc between them was empty in x; every other initially occupied site lies outside the displayed fiber, where Φ_x is injective and avoids both ports.

For completeness, a fusion between two *distinct initial packets* at any earlier prefix would make their atom sites equal at every later prefix: site maps never split a packet. Their Φ_x-images would then coincide, contradicting injectivity on the initially occupied sites just

described. This does not rule out packets already fused before the probe, nor does it need to. Thus the prefix v_x has no new fusion, and its final d creates exactly F with ordered parents $\{0\}, \{1\}$. □

For each $1 \le h \le g$, let $x^{(h)}$ retain every packet and site of x except that singleton $\{1\}$ is moved from $b = a + g$ to $a + h$; thus $x^{(g)} = x$. Choose integer lifts $\widetilde{b} = \widetilde{a} + g$ and read every subscript modulo n. For $h < g$, the actual word

$$E_{\widetilde{b}-1} E_{\widetilde{b}-2} \cdots E_{\widetilde{a}+h} \tag{5.4}$$

moves $\{1\}$ successively into holes without a fusion. For $h = g$ the word is empty. Hence all $x^{(h)}$ belong to the same generated registerable domain.

Lemma 5.3 (exact probe-coalescence fiber). For $x, y \in \mathcal{R}_n$,

$$\begin{aligned} \mathrm{Trace}(x, V_x) = \mathrm{Trace}(y, V_x) \text{ and } P(\delta_{V_x} x) = P(\delta_{V_x} y) \\ \iff \quad y = x^{(h)} \text{ for a unique } 1 \le h \le g(x). \end{aligned} \tag{5.5}$$

The case $h = g(x)$ is the identity comparison. For a distinct coalescing y, necessarily $g(y) = h < g(x)$.

Proof. Equal complete traces force the same ordered singleton parents at the last d. Let $\Psi_x = d \circ \Phi_x$ be the full site map of V_x. Since $d^{-1}(0) = \{0, 1\}$, (5.3) gives its entire zero fiber:

$$\Psi_x^{-1}(0) = \{a, a + 1, \ldots, a + g\}. \tag{5.6}$$

Every nonzero image fiber is a singleton, lying outside this interval: d fixes the other sites and Φ_x is injective on their preimages. The ordered final registration record says that, immediately before its d, source 0 is the singleton parent at port 0 and source 1 is the singleton parent at port 1. Equation (5.3) therefore forces the initial site of source 0 in y to be a, and that of source 1 to be $a + h$ for a unique $1 \le h \le g$.

Now fix an outside atom $z \notin F$. In the execution from x it starts outside $\{a, \ldots, a + g\}$, so its final site $q_z = \Psi_x(\mathrm{site}_{P_x}(z))$ is nonzero. Equal final placed partitions make its final site from y the same q_z. By atomwise execution (5.1), its initial site in y lies in $\Psi_x^{-1}(q_z)$; that fiber consists solely of its initial site in x. This holds for each outside atom, even when several already share one initial packet. Along with the recovered sites of sources $0, 1$, atom-site recovery gives exactly $y = x^{(h)}$, with unique h.

Conversely, take $y = x^{(h)}$. Its occupied sites consist of a, the new site $a + h$ of singleton $\{1\}$, and the unchanged outside sites. By (5.3), Φ_x sends these to port 0, port 1, and pairwise distinct sites away from both ports, respectively. Hence no two distinct packets of y can fuse in the prefix: if they did, they would still have the same site after v_x. The same reasoning applied above shows this for x. Both complete traces are therefore null before the last letter, where the same ordered singleton parents register F. Finally (5.6) sends both possible initial sites b and $a + h$ of source 1 to site 0; every other atom starts at the same site in x and y. Their final atom-site functions, hence their complete placed partitions, are equal. □

Example (six-site gap probe). Let $n = 6$ and take the reachable unregistered placements

$$\begin{aligned} x: &\quad 0:\{0\}, \quad 3:\{1\}, \quad 4:\{2,3\}, \quad 5:\{4,5\}, \\ y = x^{(2)}: &\quad 0:\{0\}, \quad 2:\{1\}, \quad 4:\{2,3\}, \quad 5:\{4,5\}. \end{aligned}$$

The omitted sites are holes; the actual fusion-free word E_2 moves x to y. Their gaps are 3 and 2. The probe $V_x = p^4(dp)^2 d$ maps source 0 to port 0, either initial site 2 or 3 of source 1 to port 1, and the two outside packets to sites $4, 5$ before its final d. Both runs have null output until the same registration of F, and both finish at $0:\{0,1\},\ 4:\{2,3\},\ 5:\{4,5\}$. In

the other direction, $V_y = p^5(dp)d$ has pre-final port-1 fiber $\{1, 2\}$, which excludes source 1's site 3 in x. Thus y registers under V_y, whereas x emits only null. Probing from the smaller gap prevents this coalescence.

Corollary 5.4 (pre-registration separation). Distinct states in $\mathcal{R}_n$ have a common distinguishing raw suffix of length at most $6n - 10$.

Proof. Choose a state of smaller gap as x, so $g(x) \le g(y)$. If the traces under V_x differ, stop. Otherwise (5.5) rules out equal final placements: a distinct coalescing y would have strictly smaller gap. Both endpoints registered exactly F on the final letter and are nonterminal, because $n \ge 3$ leaves at least one outside source atom. Corollary 4.4 supplies a word W_q separating their endpoint placements. For the *same* concatenated word from either initial state,

$$\mathrm{Trace}(z, V_x W_q) = \mathrm{Trace}(z, V_x) +\!+ \mathrm{Trace}(\delta_{V_x} z, W_q).$$

Thus $V_x W_q$ separates x, y, and its length is at most $(3n - 6) + (3n - 4) = 6n - 10$. □

6 The exact quotient and a uniform distinguishing depth

Theorem 1 (future-event minimal quotient). For $n \ge 3$ and reachable states $x, y \in X_n$,

$$x \equiv_{\text{future}} y \quad \Longleftrightarrow \quad \begin{cases} x, y \in \mathcal{Z}_n, & \text{or} \\ x, y \in \mathcal{R}_n \text{ and } P_x = P_y, & \text{or} \\ x, y \text{ are registered nonterminal and } P_x = P_y. \end{cases} \tag{6.1}$$

Consequently the minimal future-event machine has $Q_n^{\text{future}} = 1 + R_n + A_n$ states.

Proof. All states in $\mathcal{Z}_n$ emit null on every future letter: an unregistered state unable to register can never acquire an origin, and a rank-one registered state cannot absorb another packet. Equal current placements within $\mathcal{R}_n$ have the same unregistered state and deterministic continuation. Equal current placements in the registered nonterminal region also have identical future outputs: a selected event depends on the current packets and the carrier containing F, not on the earlier origin or chain, neither of which is re-emitted. These observations show that the stated classification defines a right congruence.

Distinct registered nonterminal placements are separated by Corollary 4.4; distinct members of $\mathcal{R}_n$ by Corollary 5.4. For cross-region pairs, V_x emits a registration event from $x \in \mathcal{R}_n$, while a sink emits only null and an already registered state can emit no *new* registration event. A registered nonterminal state has a selected absorption along some W_q, while a sink emits only null. Registration and absorption have distinct output tags. Thus no two different keys are equivalent. □

Theorem 2 (constructive uniform separation). If reachable states x, y satisfy $x \not\equiv_{\text{future}} y$, they have a common distinguishing suffix of length at most $6n - 10$. It can be chosen as a V_x, a W_q, or a concatenation $V_x W_q$. In particular, agreement on all future-event traces of words of length at most $6n - 10$ implies equivalence.

Proof. The cross-region and registered cases in Theorem 1 use one probe. Corollary 5.4 handles distinct registerable pre-registration placements and is the only case requiring a concatenation. All words are actual words over $\{p, d\}$, with nonnegative rotation exponents. The probe bounds (4.1) and (5.2) give the stated cap. □

Lemma 6.1 (event-to-summary prefix detection). Let x, y be reachable unregistered states. If a common word w gives different complete selected-event traces, then at some prefix v of w the full-summary Moore outputs of $\delta_v x$ and $\delta_v y$ differ.

Proof. Both starts have origin $\perp$ and empty selected chain. Up to the first letter at which the traces differ, equal selected-event outputs make the origin and chain updates identical. At that letter, differing complete records, or a record on only one side, make the updated origins or chains differ: registration sets the origin, and each later selected absorption appends its record. If just one side is registered, its displayed summary differs from $\perp$; if both are registered, their displayed summaries include the differing origin or chain. Either way, the prefix ending at that letter distinguishes the Moore outputs. □

For the full-summary observer, the empty word separates unequal registered summaries. Corollary 5.4 and Lemma 6.1 separate distinct members of $\mathcal{R}_n$ under this observer as well. All permanently unregistered states display $\perp$ under every continuation. A member of $\mathcal{R}_n$ is distinguished from them by a registration probe; a registered state is displayed distinctly from $\perp$ already at the empty word. Hence

$$\boxed{Q_n^{\text{full}} = s_n + R_n + 1.} \tag{6.2}$$

The registered rank-one states are *not* merged into the full-summary sink: their current summaries are displayed. The distinction between (6.1) and (6.2) is entirely due to the declared observation contracts.

7 Counts, asymptotics, and the explicit coalescence graph

Substituting Proposition 3.3 into Theorem 1 gives

$$\boxed{Q_n^{\text{future}} = 1 + n\binom{2n-4}{n-3} + 2(2n-3)\binom{2n-4}{n-2} - n(n-1).} \tag{7.1}$$

For the first growth base, put $m = n-2$ and write

$$R_n = n\binom{2m}{m-1} = n\frac{m}{m+1}\binom{2m}{m}, \qquad A_n = 2(2m+1)\binom{2m}{m} - n(n-1).$$

Since $\binom{2m}{m} \sim 4^m/\sqrt{\pi m}$, the polynomial term in A_n is negligible and the two contributions are, separately,

$$R_n \sim \frac{1}{16\sqrt{\pi}}\sqrt{n}\,4^n, \qquad A_n \sim \frac{1}{4\sqrt{\pi}}\sqrt{n}\,4^n.$$

Their sum gives

$$Q_n^{\text{future}} \sim \frac{5}{16\sqrt{\pi}}\sqrt{n}\,4^n. \tag{7.2}$$

The upstream complete-summary theorem (Chen, 2026a) gives, with $m = n-2$,

$$s_n = n\left[\binom{2m}{m} + 2\sum_{b=1}^{m} 3^{b-1}\binom{2m-b}{m-b}\right] \tag{7.3}$$

The 9/2 base and its constant can be read directly from this exact formula. Let B_m be the bracket in (7.3), defined also at $m = 0$ by that expression. For $m \geq 1$ it equals $s_{m+2}/(m+2)$. Set $C(z) = (1 - \sqrt{1-4z})/(2z)$. The standard identity $\sum_{j\geq 0} \binom{2j+b}{j} z^j = C(z)^b/\sqrt{1-4z}$ evaluates each fixed-b part of the sum.

$$\begin{aligned} \sum_{m\geq 0} B_m z^m &= \frac{1}{\sqrt{1-4z}} \left(1 + 2 \sum_{b\geq 1} 3^{b-1} (zC(z))^b \right) \\ &= \frac{1}{\sqrt{1-4z}} \frac{1 - zC(z)}{1 - 3zC(z)}. \end{aligned} \tag{7.3a}$$

Inside $|z| < 1/4$, the denominator has a single zero at $z = 2/9$. At that point $\sqrt{1-4z} = 1/3$ and $zC(z) = 1/3$, so the principal part of (7.3a) is $(1 - \frac{9}{2}z)^{-1}$. Subtracting it leaves a function analytic on $|z| < 1/4$; Cauchy's estimate on any circle with $2/9 < |z| < 1/4$ shows $B_m \sim (9/2)^m$. Hence

$$s_n \sim n \left(\frac{9}{2}\right)^{n-2} = \frac{4n}{81} \left(\frac{9}{2}\right)^n. \tag{7.3b}$$

It follows from (6.2) and the smaller exponential base of R_n that

$$Q_n^{\text{full}} \sim s_n, \qquad \boxed{\frac{Q_n^{\text{full}}}{Q_n^{\text{future}}} \sim \frac{64\sqrt{\pi}}{405} \sqrt{n} \left(\frac{9}{8}\right)^n.} \tag{7.4}$$

Thus the two minimal right congruences on one generated state space have different growth bases. This is not a lower bound on an arbitrary implementation of either observation contract.

The gap probe also has a precise directed coalescence graph. Draw $x \to y$ for distinct $x, y \in \mathcal{R}_n$ when their V_x traces and final placements agree. By Lemma 5.3,

$$x \to y \quad \Longleftrightarrow \quad y = x^{(h)} \text{ for a unique } 1 \leq h < g(x). \tag{7.5}$$

It is acyclic because g strictly decreases. Let $D_n(g)$ count registerable placements of gap g and C_n count the directed edges of this *specified probe* graph. To count $D_n(g)$, choose the site of $\{0\}$ in n ways. The $g-1$ sites before $\{1\}$ are holes; after $\{1\}$ there are $n-g-1$ available sites for t outside packets. Cutting the $n-2$ outside atoms into t nonempty source intervals gives $\binom{n-3}{t-1}$ choices, and placing them in positive order gives $\binom{n-g-1}{t}$ choices. Vandermonde's identity yields

$$D_n(g) = n \sum_{t=1}^{n-g-1} \binom{n-3}{t-1} \binom{n-g-1}{t} = n \binom{2n-g-4}{n-g-2}, \qquad 1 \leq g \leq n-2. \tag{7.6}$$

Each source vertex of gap g has exactly $g-1$ distinct outgoing comparison states. Therefore

$$\begin{aligned} C_n &= \sum_{g=1}^{n-2} (g-1) D_n(g) \\ &= n \binom{2n-4}{n-4} = \frac{n-3}{n} R_n, \end{aligned} \tag{7.7}$$

with $C_3 = 0$.

For the weighted step, put $m = n - 2$ and $j = m - g$. Hockey-stick summation and $j\binom{m+j}{j} = (m+1)\binom{m+j}{j-1}$ give, for $m \geq 2$,

$$\begin{aligned} \frac{C_n}{n} &= \sum_{j=0}^{m-1} (m-1-j)\binom{m+j}{j} \\ &= (m-1)\binom{2m}{m-1} - (m+1)\binom{2m}{m-2} \\ &= \binom{2m}{m-2}. \end{aligned} \tag{7.7a}$$

Because $R_n/n = \binom{2m}{m-1}$ and $\binom{2m}{m-2}/\binom{2m}{m-1} = (m-1)/(m+2)$, this is precisely the ratio $(n-3)/n$ in (7.7). Equivalently, the mean nonidentity outdegree of a uniformly chosen $x \in \mathcal{R}_n$ is $(n-3)/n < 1$. For each fixed positive integer g,

$$\frac{D_n(1)}{R_n} = \frac{n-1}{2n-4}, \qquad \frac{D_n(g+1)}{D_n(g)} = \frac{n-g-2}{2n-g-4},$$

so $D_n(g)/R_n \to 2^{-g}$. Since $\sum_{g \geq 1} 2^{-g} = 1$, the gap distribution converges to the geometric law of parameter $1/2$. These statements concern V_x and do not classify coalescence under other words.

8 Finite-state checks and scope

The accompanying finite-state control package computes both right congruences for $n = 3, 4, 5$, verifies the placement counts and normal-form classification, and executes a distinguishing word for one representative of every pair of distinct quotient keys. An independently implemented packet update checks every raw transition. These computations are finite consistency checks; the all-parameter results follow from the proofs above. The complete-summary count s_n is imported from (Chen, 2026a).

The theorem concerns the fixed adjacent pair, the complete selected-event alphabet, and unrestricted future words. It does not identify a quotient for truncated event records, other selected source sets, arbitrary defects, shortest distinguishing words, reset bounds, or typed transfer semantics.

9 Conclusion

The future-event and full-summary observers induce different minimal right congruences on the same generated packet system, with respective exponential growth bases 4 and $9/2$. The exact gap-probe coalescence fiber identifies which registerable starts share its terminal event and placement; together with the registered sweep, it yields distinguishing words of length at most $6n - 10$ for every inequivalent future-event pair. Thus the minimal-state cost of retaining complete registered histories separates from the state sufficient to predict future selected events, even as pairwise separation remains constructive and linear in n.

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